UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

© Hak Cipta Universiti Teknologi MARA

PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Computer generations and history

The integrated circuit is an important milestone that marks the achievement in the history of computer development.
a) In which computer generation that the integrated circuil was first introduced?
b) State the name of the inventor of the integrated circuit.
c) What are the THREE (3) main components of integrated circuits?

Source: CSC429 (20).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Data representation and number systems

Convert 12.5 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

Answer: 12.510 = 0x41480000 (sign=0, exponent=10000010, fraction=10010000000000000000000)

- *1 mark for the sign bit.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the correct 32-bit grouping.*
- *1 mark for the final hexadecimal value.*

QUESTION A3 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Parallel and multiprocessor architecture

a) List TWO (2) advantages of SMP organization.
b) Define cluster organization.
c) Give ONE (1) benefit of cluster organization.

Source: CSC429 (7).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A5 (10 marks)

Topic: Input/output and interrupts

Keyboard is the most common means for computer interaction.

© Hak Cipta Universiti Teknologi MARA

a) Explain TWO (2) types of keyboard characters.
b) Describe THREE (3) categories of input and output module. Give an appropriate example for each category.

Source: CSC429 (2).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Data representation and number systems

Answer both parts. Show your complete workflow.
a) Convert 291010 to hexadecimal.
b) Convert 29016 to decimal.

a) Convert 291010 to hexadecimal.

Answer: 291010 = B5E16

- *1 mark for repeated division by 16.*
- *1 mark for correct remainders.*
- *1 mark for reading remainders in reverse order.*
- *1 mark for the correct hexadecimal digits.*
- *1 mark for the base subscript.*

b) Convert 29016 to decimal.

Answer: 29016 = 65610

- *1 mark for place values 16^2, 16^1 and 16^0.*
- *1 mark for correct digit values.*
- *1 mark for correct products.*
- *1 mark for correct addition.*
- *1 mark for the final decimal value.*

QUESTION A7 (10 marks)

Topic: Memory hierarchy and cache

a) List FOUR (4) types of access method for storage device with ONE (1) example for each of them.
b) Calculate the total of memory location using 11 address bit.
c) Calculate the address bits required for 2MB of RAM.

Source: CSC429 (10).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A8 (10 marks)

Topic: Computer architecture and CPU organisation

Given that the size of the data bus is 32 bits, the size of the address bus is 8 bits, and the size of the control bus is 4 bits.
a) Determine the size of the word and provide relevant justification.
b) Determine the size of the register and provide relevant justification.
c) Discuss what will happen if system interconnection is blocked from the perspective of the machine cycle.
d) Elaborate the role of CPU interconnection.

Source: CSC429 (17).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Input/output and interrupts

Question 7 is based on Figure 2.
lnterrupt handler W
User program
--'-'j

© Hak Cipta Universiti Teknologi MARA

lnterrupt handler X
-----:
lnterrupt handler Y
lnterrupt handler Z
t
Figure 2: Nested interrupts
Define intenupt.
a)
ldentify the intemrpt that has the highest priority with relevant justification
b)
ldentify the interrupt that has the lowest priority with relevant justification
c)
Rank the sequence of the intem.rpt process completed ff the computer system applies the
d)
defined priority concept to handle multiple intemrpts.
mputer system aPPlies the
Rank the sequence of the interrupt process completed if the co
e)
disable nested interrupt concept to handle multiple interrupts'

Source: CSC429 (21).pdf, page(s) 3, 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Computer architecture and CPU organisation

a)
List TWO (2) buses types that are available in the computer system.
b)
Differentiate TWO (2) characteristics of system interconnection
and CPU
interconnection.
c)
Explain the different types of data transfer timing.

Source: CSC429 (15).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

© Hak Cipta Universiti Teknologi MARA

PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.

y = A -
2C
* D
B + A2

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three-address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

Three-address	One-address	Two-address
LOADp		
MOV (p,q)		
ADD (p.q.r)		
STORp		
ADD (p,q)		
SUB (p,q,r)		
ADDp		
SUB (p,q)		
MUL (p.q.r)		
SUBp		
MUL (p,q)		
DIV(p,q,r)		
MULp		
DIV(p,q)		
DIVp		

Source: CSC429 (18).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

© Hak Cipta Universiti Teknologi MARA